CONSTRUCTIBILITY OF A SHEAF ON THE SMALL ETALE SITE IS A ZARISKI LOCAL PROPERTY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the reformulation, the restriction to affine opens and the application of the criterion are all inside the counted span.

Editorial note. The label lemma-constructible-local occurs in no other staged Stacks chapter, so no cross-chapter restatement had to be weighed for this card.

Editorial note. The single labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer to the same chapter.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Constructible sheaf of sets, abelian groups or modules on X_etale. Same chapter, the definition of constructibility by a partition of X into locally closed constructible strata on which the sheaf is locally constant. Supplies the property shown to be local on X.
- Restriction of a constructible sheaf to an open subscheme. Same chapter, the immediate fact that the restriction of a constructible sheaf to an open subscheme is constructible. Supplies the constructibility of the restrictions to the members of a refinement of the given Zariski covering.
- Zariski local nature of constructibility on a quasi-compact quasi-separated scheme. Same chapter, the criterion producing a partition of an affine scheme from constructible restrictions on the members of a finite open covering. Supplies the final step, which assembles the local partitions into a global one.

Source reference. Original lemma 095E: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 11717–11717. Same-chapter navigation pointer invoked at proof lines 11801-11802, where the author uses the criterion for constructibility on a quasi-compact quasi-separated scheme in terms of finitely many strata. The pointer only resolves that reference for the reader; the cited lemma is not reproduced inside the counted span, and no bibliography citation is involved.

Lemma 0.1. *Let X be a scheme. Checking constructibility of a sheaf of sets, abelian groups, Λ-modules (with Λ Noetherian) can be done Zariski locally on X.*

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Proof. The statement means if $X = \bigcup U_i$ is an open covering such that $\mathcal{F}|_{U_i}$ is constructible, then $\mathcal{F}$ is constructible. If $U \subset X$ is affine open, then $U = \bigcup U \cap U_i$ and $\mathcal{F}|_{U \cap U_i}$ is constructible (it is trivial that the restriction of a constructible sheaf to an open is constructible). It follows from Lemma 095E that $\mathcal{F}|_U$ is constructible, i.e., a suitable partition of U exists. $\square$